

ECE 220 Midterm #1
Thursday, October 6, 2023

Name: Solutions

- 5 problems total.
- Harder / longer problems are towards the back. Allot time appropriately.
- 90 minutes (5:45 – 7:15 pm) – this may differ for makeups or accommodated testing
- You may use one 8.5" x 11" (both sides) sheet of notes
- You may use a non-graphing calculator
- There are equation sheets at the back of the test. You may tear these off if you wish, and you do not have to turn them in with your test.
- You may use the backs of pages if you need extra space for calculations. If you do so, please clearly indicate that your work continues on the back, and be sure to label your work with the corresponding problem.

Good luck!

Problem 1: 20

Problem 2: 20

Problem 3: 20

Problem 4: 20

Problem 5: 20

Total: 100 / 100

Problem 1:

- a) Calculate curl and divergence of the following vector field. State whether the field is conservative, solenoidal, both, or neither.

$$\vec{A} = \hat{r} 4 \exp(-5r) / r$$

+3 $\boxed{\nabla \times \vec{A} = 0}$ because only $\hat{r}$ component & depends on r only

$$\nabla \cdot \vec{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) = \frac{1}{r} \frac{\partial}{\partial r} (4e^{-5r}) = \frac{1}{r} (-20e^{-5r})$$

+3 $\Rightarrow \boxed{\nabla \cdot \vec{A} = -\frac{20}{r} e^{-5r}} \neq 0$

+2

Field is conservative

- b) Calculate curl and divergence of the following vector field. State whether the field is conservative, solenoidal, both, or neither.

$$\vec{A} = \hat{x}(5x - 2yz) + \hat{y}(10y + 3xz) + \hat{z}(-15z + 4xy)$$

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \hat{x}(4x - 3x) + \hat{y}(-2y - 4y) + \hat{z}(3z + 2z)$$

+3 $\Rightarrow \boxed{\nabla \times \vec{A} = \hat{x}x - \hat{y}6y + \hat{z}5z} \neq 0$

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 5 + 10 - 15 \Rightarrow \boxed{\nabla \cdot \vec{A} = 0} \quad +3$$

Field is solenoidal +2

- c) Some vector field $\vec{A}$ is related to a scalar field f as $\vec{A} = \nabla f$. What is the value of $\nabla \times \vec{A}$? Explain briefly (1-2 sentences).

+2 $\boxed{\nabla \times \vec{A} = \nabla \times (\nabla f) = 0}$

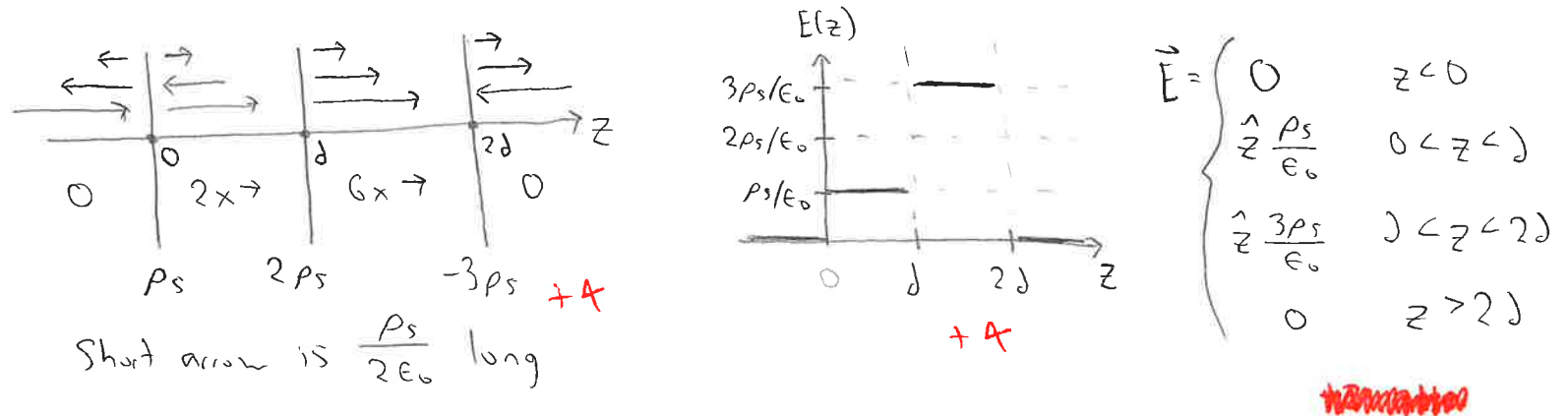
Vector fields that result from the gradient of a scalar field are always conservative. +2

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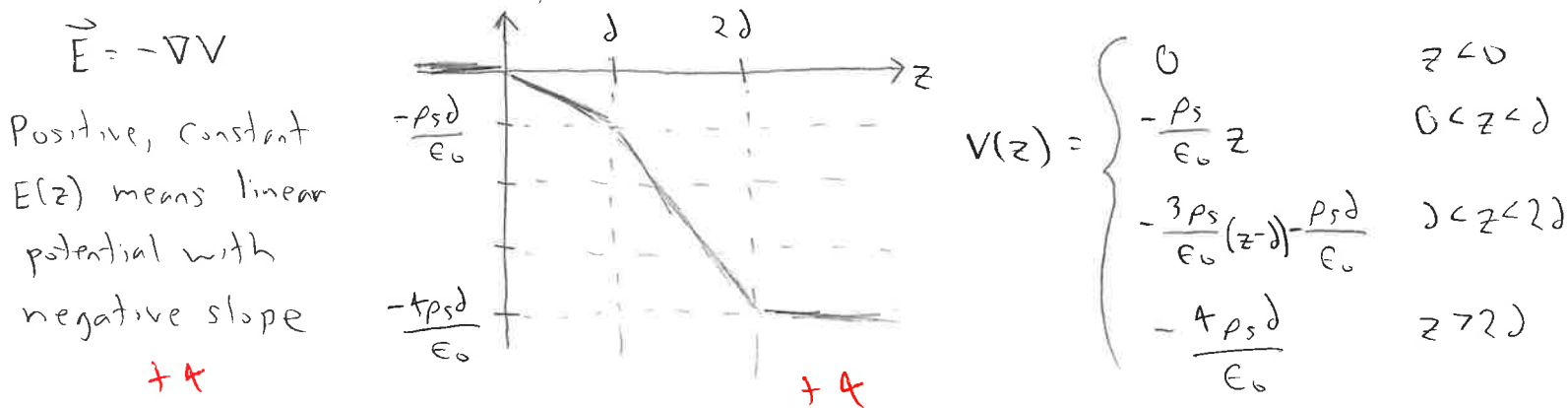
Problem 2:

Three infinite parallel planes have surface charge densities ρ_s , $2\rho_s$, $-3\rho_s$ ($\rho_s > 0$). They are normal to the z -axis and located at $z = 0$, $z = d$, and $z = 2d$, respectively. Assume the planes are in vacuum (permittivity ϵ_0).

- a) Find $\vec{E}$ everywhere in space. Graph $E(z)$ versus z . Label all important points on the axes.

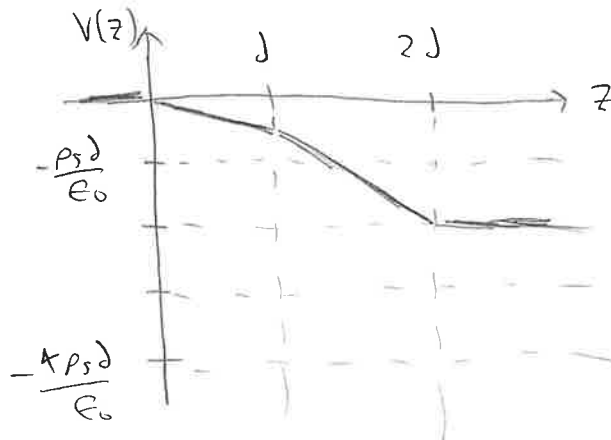


- b) Calculate and graph the electric potential $V(z)$. Assume $V(0) = 0$. Label all important points on the axes.



- c) How does the result from part (b) change if the material between the leftmost and rightmost plates is a dielectric with relative permittivity $\epsilon_r = 2$ instead of free space? Be specific about any quantitative changes.

Shape will not change but magnitude will be $\frac{1}{\epsilon_r} = \frac{1}{2}$



20

Problem 3:

Consider two infinitely long wires, both parallel to the y-axis, one located at $x = -a$ and $z = 0$, the other at $x = 0$ and $z = a$ (see picture). The two wires have the same line charge density ρ_l . The material throughout is vacuum.

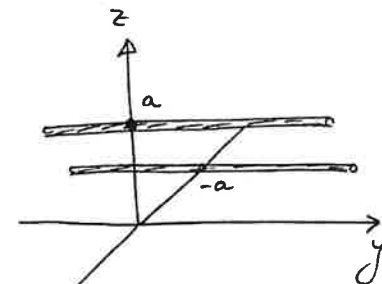
+5

- a) Write down the electric field vector $\vec{E}$ at any point P on the y-axis (i.e., for points that have $x = 0$ and $z = 0$).

$$\vec{E} = \hat{x} \frac{\rho_l}{2\pi\epsilon_0 a}$$

$$\vec{E} = -\hat{z} \frac{\rho_l}{2\pi\epsilon_0 a}$$

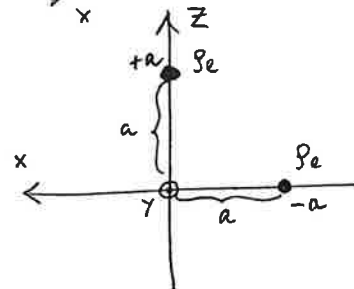
$$\vec{E} = \hat{x} \frac{\rho_l}{2\pi\epsilon_0 a} - \hat{z} \frac{\rho_l}{2\pi\epsilon_0 a}$$



- b) If charge Q is placed at the coordinate origin, what is the force $\vec{F}$ (magnitude and direction) it feels from the two wires? Remember that the force on charge Q is related to the electric field at that point as $\vec{F} = Q\vec{E}$.

+2

$$\vec{F} = Q\vec{E} = \frac{Q\rho_l}{2\pi\epsilon_0 a} (\hat{x} - \hat{z})$$



- c) How much energy would it cost (i.e., how much work is needed) to move charge Q from the coordinate origin to any other point along the y-axis? Hint: work needed to move a charge Q from point A to point B is related to the potential by $W = Q(V_B - V_A) = -Q \int_A^B \vec{E} \cdot d\vec{l}$

+5

$\vec{E}$ and $\vec{F}$ are $\perp$ to $\hat{y}$, so $\vec{E} \cdot d\vec{l} = 0$,

so no work is needed

- d) How much energy would it cost (i.e., how much work is needed) to move charge Q from the origin to a point $(a, 0, 0)$? For reference, the potential difference between points at normal distances r_1 and r_2 from a wire line charge density ρ_l can be written as

+8

$$V(r_2) - V(r_1) = \frac{\rho_l}{2\pi\epsilon} \ln\left(\frac{r_1}{r_2}\right)$$

$$W = Q(V_B - V_A) = Q \left(\frac{\rho_l}{2\pi\epsilon_0} \ln\left(\frac{a}{2a}\right) + \frac{\rho_l}{2\pi\epsilon_0} \ln\left(\frac{a}{a\sqrt{2}}\right) \right)$$

wire at $x = -a$ wire at $z = a$

$$W = \frac{Q\rho_l}{2\pi\epsilon_0} \ln\left(\frac{a^2}{2a^2\sqrt{2}}\right) = \frac{Q\rho_l}{2\pi\epsilon_0} \ln\left(\frac{1}{2\sqrt{2}}\right)$$

$$= -\frac{3}{4} \frac{Q\rho_l}{\pi\epsilon_0} \ln 2$$

External work not needed
because $W < 0$

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Problem 4:

An infinitely long cylinder with relative permittivity $\epsilon_r > 1$ and radius a is charged with a volume charge density

$$\rho_v = \rho_{v0} \frac{r^2}{a^2}, \quad r < a$$

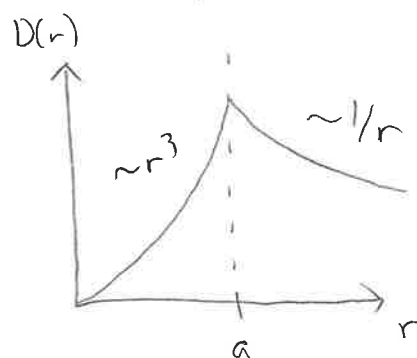
There is no charge outside the cylinder. Free space (air) surrounds the cylinder.

a) Use Gauss's law to calculate $\vec{D}$ everywhere in space. Graph $D(r)$ vs r .

$r < a$: $2\pi r h D(r) = \int \rho_v dV$ +2

$$r D(r) = \int_0^r \rho_{v0} \left(\frac{r'^2}{a^2} \right) r' dr'$$

$$= \rho_{v0} \frac{r^4}{4a^2} \Big|_0^r \Rightarrow D(r) = \rho_{v0} \frac{r^3}{4a^2}$$
 +1



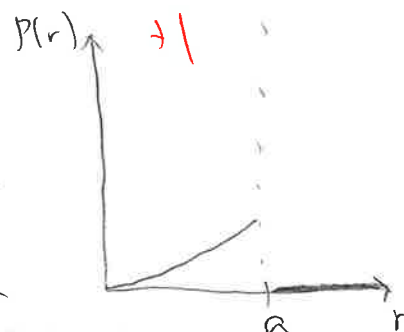
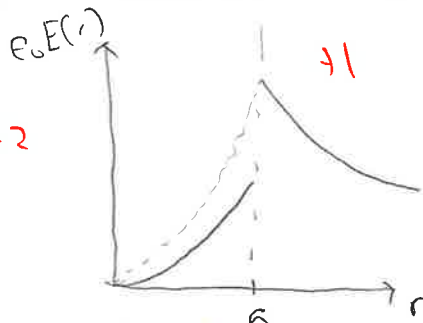
$r > a$: $2\pi r h D(r) = 2\pi h \rho_{v0} \left(\frac{a^2}{4} \right)$ +2

$$\Rightarrow D(r) = \rho_{v0} \frac{a^2}{4r}$$
 +1

b) Calculate $E(r)$ and graph $\epsilon_0 E(r)$ vs r everywhere. Graph $P(r)$ vs r everywhere.

$r < a$: $E(r) = \frac{D(r)}{\epsilon_r \epsilon_0} = \frac{\rho_{v0}}{\epsilon_r \epsilon_0} \cdot \frac{r^3}{4a^2}$

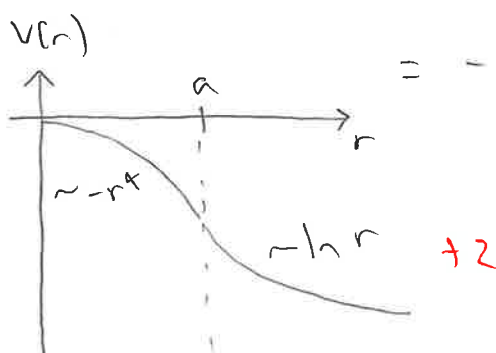
$r > a$: $E(r) = \frac{D(r)}{\epsilon_0} = \frac{\rho_{v0}}{\epsilon_0} \cdot \frac{a^2}{4r}$



c) Calculate and graph the electric potential $V(r)$, assuming it is zero on the cylinder axis.

$r < a$: $V(r) - V(0) = - \int_0^r E(r') dr' = - \frac{\rho_{v0}}{\epsilon_r \epsilon_0} \frac{r^4}{16a^2}$ +2 +1

$r > a$: $V(r) - V(0) = V(a) + - \int_a^r \frac{\rho_{v0} a^2}{4r' \epsilon_0} dr'$ +2



$$= - \frac{\rho_{v0}}{\epsilon_r \epsilon_0} \frac{a^2}{16} - \frac{\rho_{v0} a^2}{4\epsilon_0} \ln \left(\frac{r}{a} \right) = - \frac{\rho_{v0} a^2}{4\epsilon_0} \left(\frac{1}{4\epsilon_r} + \ln \left(\frac{r}{a} \right) \right)$$

$$= - \frac{\rho_{v0} a^2}{4\epsilon_0} \left(\frac{1}{4\epsilon_r} + \ln \frac{r}{a} \right)$$
 +1

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Problem 5:

Charge Q has been placed on a metallic sphere of radius a (i.e., the sphere is made of an excellent conductor). The surrounding medium is air (relative permittivity $\epsilon_r = 1$).

- a) Use Gauss's law to calculate the electric flux density $\vec{D}$ and electric field intensity $\vec{E}$ everywhere in space. Graph $D(R)$ vs R and $\epsilon_0 E(R)$ vs R . (Hint: The sphere is metallic. That should guide your assumptions about how the charge is distributed.)

Excellent conductor $\rightarrow$ surface charge only $\rightarrow \rho_v = 0, \rho_s = \frac{Q}{4\pi a^2}$

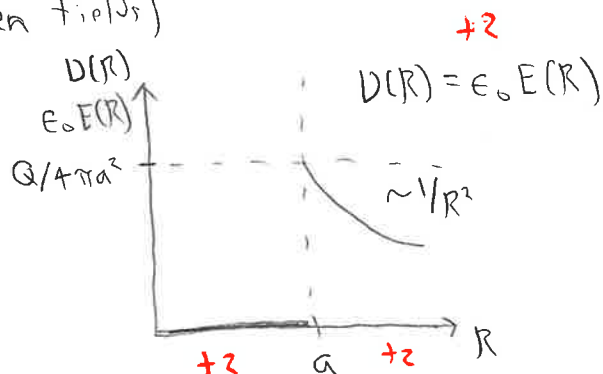
$R < a: \vec{D} = \vec{E} = 0$ (conductors screen fields)

$$R > a: 4\pi R^2 D(R) = Q$$

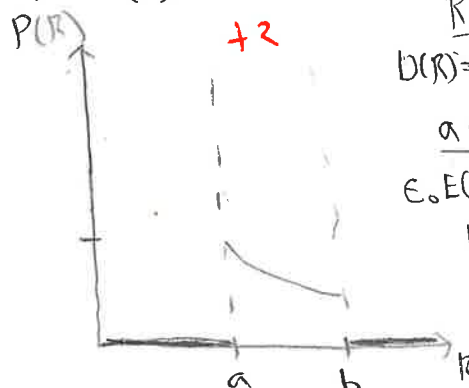
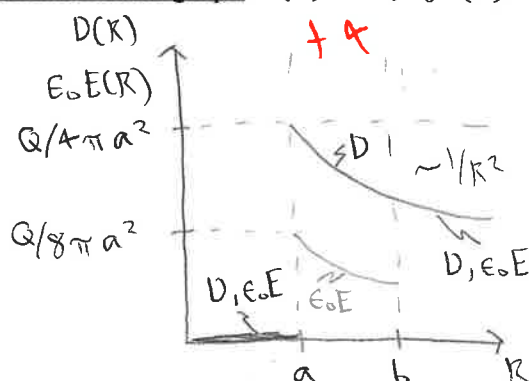
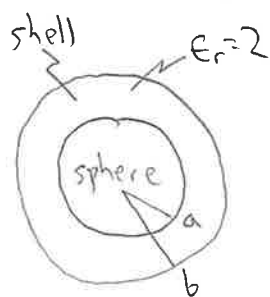
$$\Rightarrow D(R) = \frac{Q}{4\pi R^2}$$

$$E(R) = \frac{Q}{4\pi R^2 \epsilon_0}$$

$$\vec{D} = D(R) \hat{R}, \vec{E} = E(R) \hat{R}$$



- b) A thick dielectric spherical shell is added to the system (i.e., in addition to the charged sphere from before). The dielectric extends from $R = a$ to $R = b$, and has relative permittivity of $\epsilon_r = 2$. Calculate and graph $D(R)$ vs R , $\epsilon_0 E(R)$ vs R , and $P(R)$ vs R .



$$\begin{aligned} R < a: \\ D(R) &= \epsilon_0 E(R) = P(R) = 0 \\ a < R < b: \\ \epsilon_0 E(R) &= P(R) = \frac{Q}{8\pi R^2} \\ D(R) &= \frac{Q}{4\pi R^2} \\ R > b: \\ \epsilon_0 E(R) &= D(R) = \frac{Q}{4\pi R^2} \\ P(R) &= 0 \end{aligned}$$

- c) Calculate the potential difference $V(a) - V(b)$ for the setup described in part (b) (i.e., both the conductor and the dielectric shell together).

$$V(a) - V(b) = - \int_b^a E(R) dR \quad +2$$

$$= \int_a^b \frac{Q}{4\pi \epsilon_0 \epsilon_r} \cdot \frac{1}{R^2} dR = \frac{Q}{8\pi \epsilon_0} \left[-\frac{1}{R} \right]_a^b$$

$$= \frac{Q}{8\pi \epsilon_0} \left(-\frac{1}{b} + \frac{1}{a} \right)$$

$$= \frac{Q}{8\pi \epsilon_0} \cdot \frac{b-a}{ab}$$

Mathematical and Physical Models

Maxwell's Equations for Electrostatics

Name	Differential Form	Integral Form
Gauss's law	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$
Kirchhoff's law	$\nabla \times \mathbf{E} = 0$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$

Electric Field

Current density	$\mathbf{J} = \rho_v \mathbf{u}$	Point charge	$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2}$
Poisson's equation	$\nabla^2 V = -\frac{\rho_v}{\epsilon}$	Many point charges	$\mathbf{E} = \frac{1}{4\pi\epsilon} \sum_{i=1}^N \frac{q_i (\mathbf{R} - \mathbf{R}_i)}{ \mathbf{R} - \mathbf{R}_i ^3}$
Laplace's equation	$\nabla^2 V = 0$	Volume distribution	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{V'} \hat{\mathbf{R}}' \frac{\rho_v dV'}{R'^2}$
Resistance	$R = \frac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{s}}$	Surface distribution	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{S'} \hat{\mathbf{R}}' \frac{\rho_s ds'}{R'^2}$
Boundary conditions	Table 4-3	Line distribution	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{l'} \hat{\mathbf{R}}' \frac{\rho_l dl'}{R'^2}$
Capacitance	$C = \frac{\int_S \epsilon \mathbf{E} \cdot d\mathbf{s}}{-\int_l \mathbf{E} \cdot d\mathbf{l}}$	Infinite sheet of charge	$\mathbf{E} = \hat{\mathbf{z}} \frac{\rho_s}{2\epsilon_0}$
RC relation	$RC = \frac{\epsilon}{\sigma}$	Infinite line of charge	$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0} = \hat{\mathbf{r}} \frac{D_r}{\epsilon_0} = \hat{\mathbf{r}} \frac{\rho_l}{2\pi\epsilon_0 r}$
Energy density	$w_e = \frac{1}{2} \epsilon E^2$	Dipole	$\mathbf{E} = \frac{qd}{4\pi\epsilon_0 R^3} (\hat{\mathbf{R}} 2 \cos \theta + \hat{\boldsymbol{\theta}} \sin \theta)$
		Relation to V	$\mathbf{E} = -\nabla V$

$$V = \frac{1}{4\pi\epsilon} \int_{V'} \frac{\rho_v}{R'} dV' \quad (\text{volume distribution}), \quad (4.48a)$$

$$V = \frac{1}{4\pi\epsilon} \int_{S'} \frac{\rho_s}{R'} ds' \quad (\text{surface distribution}), \quad (4.48b)$$

$$V = \frac{1}{4\pi\epsilon} \int_{l'} \frac{\rho_l}{R'} dl' \quad (\text{line distribution}). \quad (4.48c)$$

Here, $\mathbf{R}'$ denotes the vector pointing from source at $\mathbf{R}_0$ to point of observation at $\mathbf{R}$, i.e., $\mathbf{R}' = \mathbf{R} - \mathbf{R}_0$

Charge-Current Continuity

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

Conductor Charge Dissipation

$$\rho_v(t) = \rho_{v0} e^{-(\sigma/\epsilon)t} = \rho_{v0} e^{-t/\tau_c}$$

Current Density

$$\text{Conduction} \quad \mathbf{J}_c = \sigma \mathbf{E}$$

Integral form

$$\oint \vec{J} \cdot d\vec{S} = -\frac{d}{dt} Q_{\text{inside}}$$

Table 3-1 Summary of vector relations.

	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Coordinate variables	x, y, z	r, ϕ, z	R, θ, ϕ
Vector representation $\mathbf{A} =$	$\hat{x}A_x + \hat{y}A_y + \hat{z}A_z$	$\hat{r}A_r + \hat{\phi}A_\phi + \hat{z}A_z$	$\hat{R}A_R + \hat{\theta}A_\theta + \hat{\phi}A_\phi$
Magnitude of $\mathbf{A}$ $ \mathbf{A} =$	$\sqrt{A_x^2 + A_y^2 + A_z^2}$	$\sqrt{A_r^2 + A_\phi^2 + A_z^2}$	$\sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$
Position vector $\overrightarrow{OP_1} =$	$\hat{x}x_1 + \hat{y}y_1 + \hat{z}z_1$, for $P(x_1, y_1, z_1)$	$\hat{r}r_1 + \hat{z}z_1$, for $P(r_1, \phi_1, z_1)$	$\hat{R}R_1$, for $P(R_1, \theta_1, \phi_1)$
Base vectors properties	$\hat{x} \cdot \hat{x} = \hat{y} \cdot \hat{y} = \hat{z} \cdot \hat{z} = 1$ $\hat{x} \cdot \hat{y} = \hat{y} \cdot \hat{z} = \hat{z} \cdot \hat{x} = 0$ $\hat{x} \times \hat{y} = \hat{z}$ $\hat{y} \times \hat{z} = \hat{x}$ $\hat{z} \times \hat{x} = \hat{y}$	$\hat{r} \cdot \hat{r} = \hat{\phi} \cdot \hat{\phi} = \hat{z} \cdot \hat{z} = 1$ $\hat{r} \cdot \hat{\phi} = \hat{\phi} \cdot \hat{z} = \hat{z} \cdot \hat{r} = 0$ $\hat{r} \times \hat{\phi} = \hat{z}$ $\hat{\phi} \times \hat{z} = \hat{r}$ $\hat{z} \times \hat{r} = \hat{\phi}$	$\hat{R} \cdot \hat{R} = \hat{\theta} \cdot \hat{\theta} = \hat{\phi} \cdot \hat{\phi} = 1$ $\hat{R} \cdot \hat{\theta} = \hat{\theta} \cdot \hat{\phi} = \hat{\phi} \cdot \hat{R} = 0$ $\hat{R} \times \hat{\theta} = \hat{\phi}$ $\hat{\theta} \times \hat{\phi} = \hat{R}$ $\hat{\phi} \times \hat{R} = \hat{\theta}$
Dot product $\mathbf{A} \cdot \mathbf{B} =$	$A_x B_x + A_y B_y + A_z B_z$	$A_r B_r + A_\phi B_\phi + A_z B_z$	$A_R B_R + A_\theta B_\theta + A_\phi B_\phi$
Cross product $\mathbf{A} \times \mathbf{B} =$	$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{r} & \hat{\phi} & \hat{z} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{R} & \hat{\theta} & \hat{\phi} \\ A_R & A_\theta & A_\phi \\ B_R & B_\theta & B_\phi \end{vmatrix}$
Differential length $d\mathbf{l} =$	$\hat{x} dx + \hat{y} dy + \hat{z} dz$	$\hat{r} dr + \hat{\phi} r d\phi + \hat{z} dz$	$\hat{R} dR + \hat{\theta} R d\theta + \hat{\phi} R \sin \theta d\phi$
Differential surface areas	$ds_x = \hat{x} dy dz$ $ds_y = \hat{y} dx dz$ $ds_z = \hat{z} dx dy$	$ds_r = \hat{r} r d\phi dz$ $ds_\phi = \hat{\phi} dr dz$ $ds_z = \hat{z} r dr d\phi$	$ds_R = \hat{R} R^2 \sin \theta d\theta d\phi$ $ds_\theta = \hat{\theta} R \sin \theta dR d\phi$ $ds_\phi = \hat{\phi} R dR d\theta$
Differential volume $dV =$	$dx dy dz$	$r dr d\phi dz$	$R^2 \sin \theta dR d\theta d\phi$

Table 3-2 Coordinate transformation relations.

Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt{x^2 + y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{r} = \hat{x} \cos \phi + \hat{y} \sin \phi$ $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$ $\hat{z} = \hat{z}$	$A_r = A_x \cos \phi + A_y \sin \phi$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r \cos \phi$ $y = r \sin \phi$ $z = z$	$\hat{x} = \hat{r} \cos \phi - \hat{\phi} \sin \phi$ $\hat{y} = \hat{r} \sin \phi + \hat{\phi} \cos \phi$ $\hat{z} = \hat{z}$	$A_x = A_r \cos \phi - A_\phi \sin \phi$ $A_y = A_r \sin \phi + A_\phi \cos \phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt{x^2 + y^2 + z^2}$ $\theta = \tan^{-1}[\sqrt{x^2 + y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{R} = \hat{x} \sin \theta \cos \phi + \hat{y} \sin \theta \sin \phi + \hat{z} \cos \theta$ $\hat{\theta} = \hat{x} \cos \theta \cos \phi + \hat{y} \cos \theta \sin \phi - \hat{z} \sin \theta$ $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$	$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$ $A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$
Spherical to Cartesian	$x = R \sin \theta \cos \phi$ $y = R \sin \theta \sin \phi$ $z = R \cos \theta$	$\hat{x} = \hat{R} \sin \theta \cos \phi + \hat{\theta} \cos \theta \cos \phi - \hat{\phi} \sin \phi$ $\hat{y} = \hat{R} \sin \theta \sin \phi + \hat{\theta} \cos \theta \sin \phi + \hat{\phi} \cos \phi$ $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$	$A_x = A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi$ $A_y = A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$
Cylindrical to spherical	$R = \sqrt{r^2 + z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{R} = \hat{r} \sin \theta + \hat{z} \cos \theta$ $\hat{\theta} = \hat{r} \cos \theta - \hat{z} \sin \theta$ $\hat{\phi} = \hat{\phi}$	$A_R = A_r \sin \theta + A_z \cos \theta$ $A_\theta = A_r \cos \theta - A_z \sin \theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R \sin \theta$ $\phi = \phi$ $z = R \cos \theta$	$\hat{r} = \hat{R} \sin \theta + \hat{\theta} \cos \theta$ $\hat{\phi} = \hat{\phi}$ $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$	$A_r = A_R \sin \theta + A_\theta \cos \theta$ $A_\phi = A_\phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$

GRADIENT, DIVERGENCE, CURL, & LAPLACIAN OPERATORS CARTESIAN (RECTANGULAR) COORDINATES (x, y, z)

$$\nabla V = \hat{x} \frac{\partial V}{\partial x} + \hat{y} \frac{\partial V}{\partial y} + \hat{z} \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \hat{x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

CYLINDRICAL COORDINATES (r, ϕ, z)

$$\nabla V = \hat{r} \frac{\partial V}{\partial r} + \hat{\phi} \frac{1}{r} \frac{\partial V}{\partial \phi} + \hat{z} \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \frac{1}{r} \begin{vmatrix} \hat{r} & \hat{\phi} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & r A_\phi & A_z \end{vmatrix} = \hat{r} \left(\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) + \hat{\phi} \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) + \hat{z} \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\phi) - \frac{\partial A_r}{\partial \phi} \right]$$

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

SPHERICAL COORDINATES (R, θ, ϕ)

$$\nabla V = \hat{R} \frac{\partial V}{\partial R} + \hat{\theta} \frac{1}{R} \frac{\partial V}{\partial \theta} + \hat{\phi} \frac{1}{R \sin \theta} \frac{\partial V}{\partial \phi}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{R \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\begin{aligned} \nabla \times \mathbf{A} &= \frac{1}{R^2 \sin \theta} \begin{vmatrix} \hat{R} & \hat{\theta} & \hat{\phi} \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_R & R A_\theta & (R \sin \theta) A_\phi \end{vmatrix} \\ &= \hat{R} \frac{1}{R \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] + \hat{\theta} \frac{1}{R} \left[\frac{1}{\sin \theta} \frac{\partial A_R}{\partial \phi} - \frac{\partial}{\partial R} (R A_\phi) \right] + \hat{\phi} \frac{1}{R} \left[\frac{\partial}{\partial R} (R A_\theta) - \frac{\partial A_R}{\partial \theta} \right] \end{aligned}$$

$$\nabla^2 V = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

SOME USEFUL VECTOR IDENTITIES

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$$

Scalar (or dot) product

$$\mathbf{A} \times \mathbf{B} = \hat{n} AB \sin \theta_{AB}$$

Vector (or cross) product, $\hat{n}$ normal to plane containing $\mathbf{A}$ and $\mathbf{B}$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

$$\nabla(U + V) = \nabla U + \nabla V$$

$$\nabla(UV) = U\nabla V + V\nabla U$$

$$\nabla \cdot (\mathbf{A} + \mathbf{B}) = \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B}$$

$$\nabla \cdot (U\mathbf{A}) = U\nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla U$$

$$\nabla \times (U\mathbf{A}) = U\nabla \times \mathbf{A} + \nabla U \times \mathbf{A}$$

$$\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

$$\nabla \times \nabla V = 0$$

$$\nabla \cdot \nabla V = \nabla^2 V$$

$$\nabla \times \nabla \times \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

$$\int_V (\nabla \cdot \mathbf{A}) dV = \oint_S \mathbf{A} \cdot d\mathbf{s} \quad \text{Divergence theorem (S encloses V)}$$

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\mathbf{l} \quad \text{Stokes's theorem (S bounded by C)}$$

Mathematical and Physical Models

Distance between Two Points

$$d = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}$$

$$d = [r_2^2 + r_1^2 - 2r_1 r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2]^{1/2}$$

$$d = \{R_2^2 + R_1^2 - 2R_1 R_2 [\cos \theta_2 \cos \theta_1 + \sin \theta_1 \sin \theta_2 \cos(\phi_2 - \phi_1)]\}^{1/2}$$

Coordinate Systems Table 3-1

Coordinate Transformations Table 3-2

Vector Products

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$$

$$\mathbf{A} \times \mathbf{B} = \hat{n} AB \sin \theta_{AB}$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

Divergence Theorem

$$\int_V \nabla \cdot \mathbf{E} dV = \oint_S \mathbf{E} \cdot d\mathbf{s}$$

Vector Operators

$$\nabla T = \hat{x} \frac{\partial T}{\partial x} + \hat{y} \frac{\partial T}{\partial y} + \hat{z} \frac{\partial T}{\partial z}$$

$$\nabla \cdot \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\nabla \times \mathbf{B} = \hat{x} \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + \hat{y} \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) + \hat{z} \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

(see back cover for cylindrical and spherical coordinates)

Stokes's Theorem

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}$$